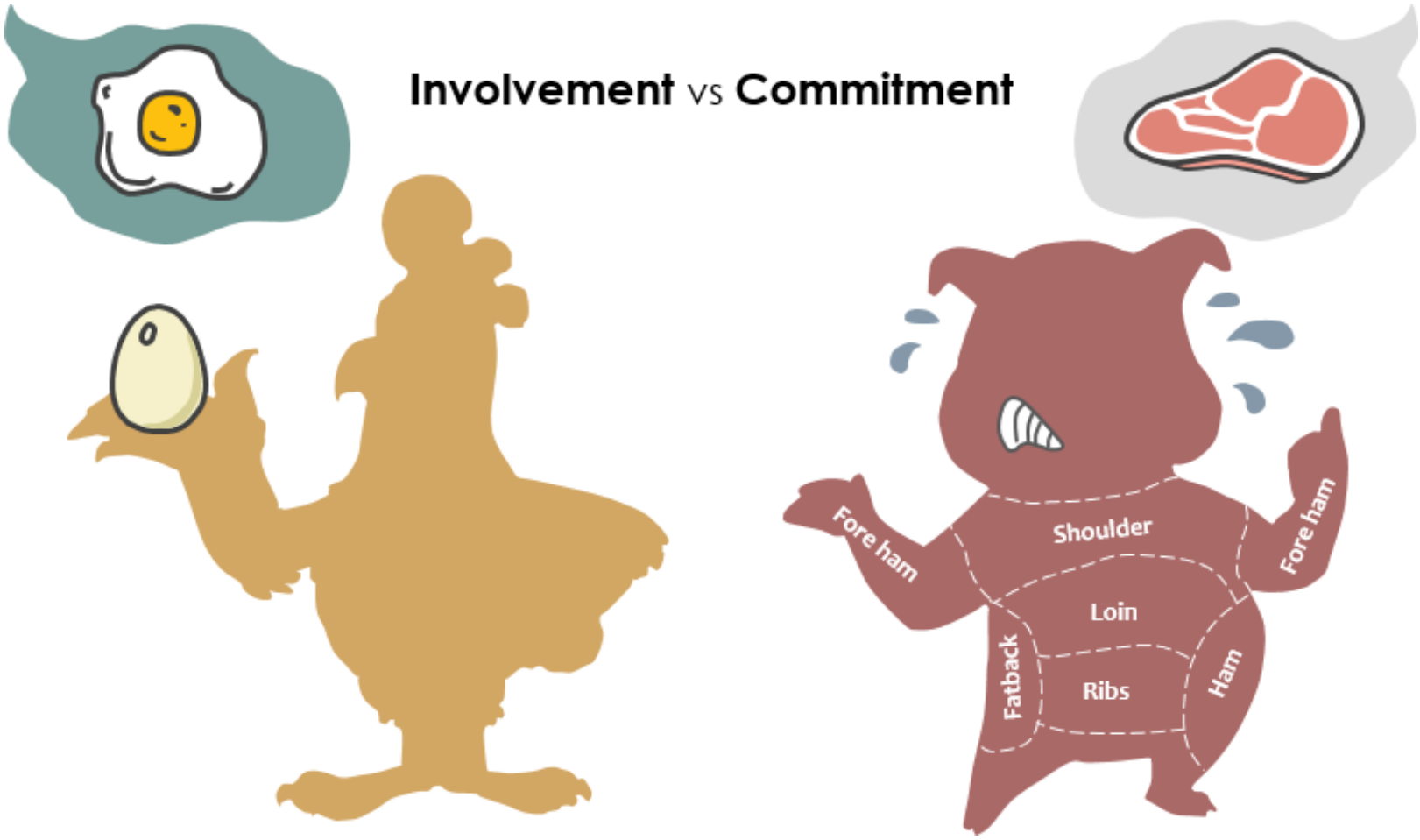


Advanced Software Development

Johan Bontes (johan@digitsolutions.nl)

CSC3003S



CSC3003S ASD09 Scrum

A. Intro to Scrum

B. Scrum flow

C. Scrum meetings

D. Process artefacts

E. Scrum risks

Scrum?

Variant of Agile

Agile: philosophy, Scrum: concrete implementation

State of Agile*: 63% of Agile projects use Scrum [1]

*) just another online survey

Warning!

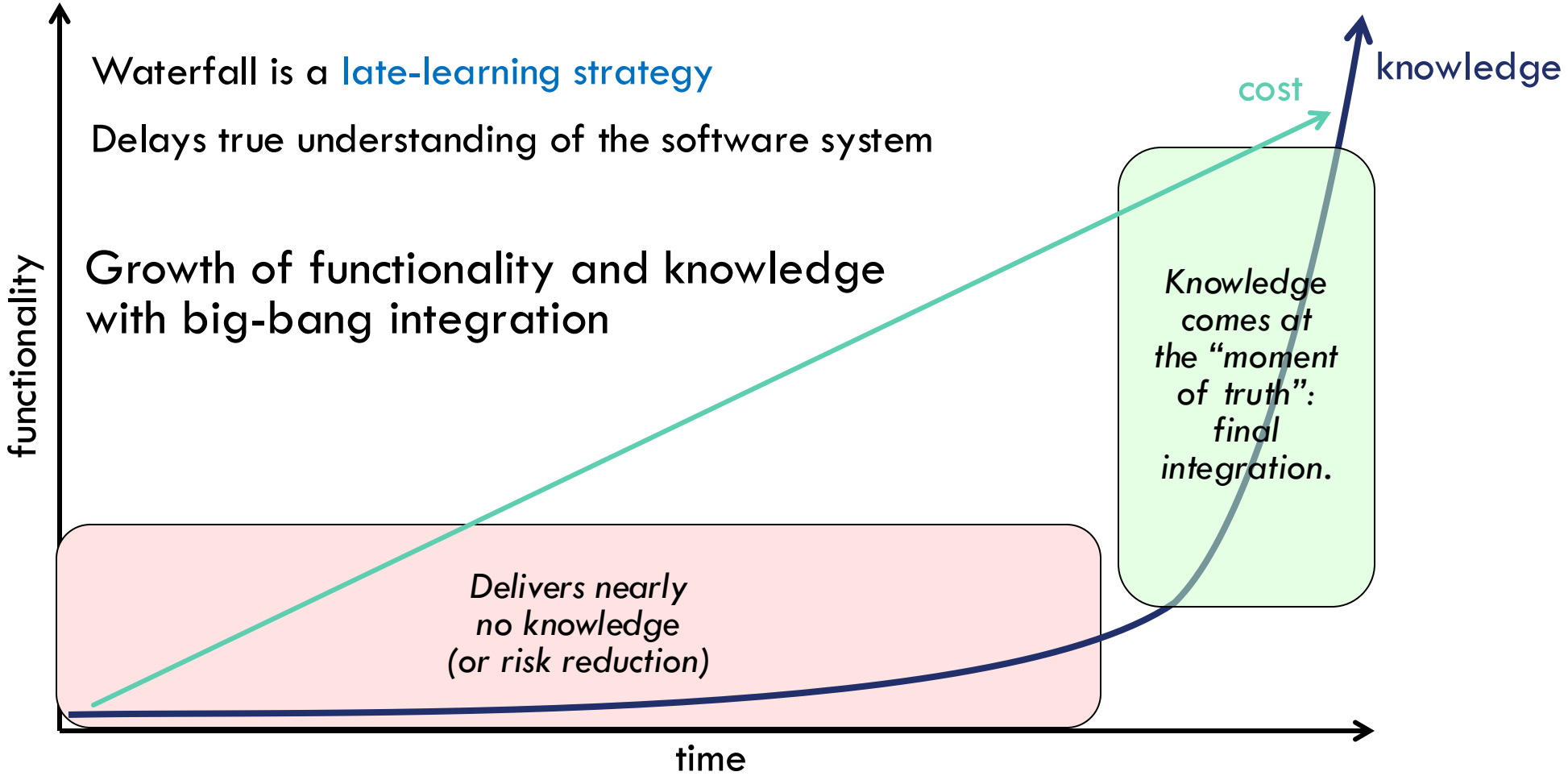
- Not a silver bullet …
  - does not compensate for lack of training or experience
    - Scrum team has defined roles
- Scrum is not a panacea for all known development ills

Waterfall: Finished functionality

Waterfall is a late-learning strategy

Delays true understanding of the software system

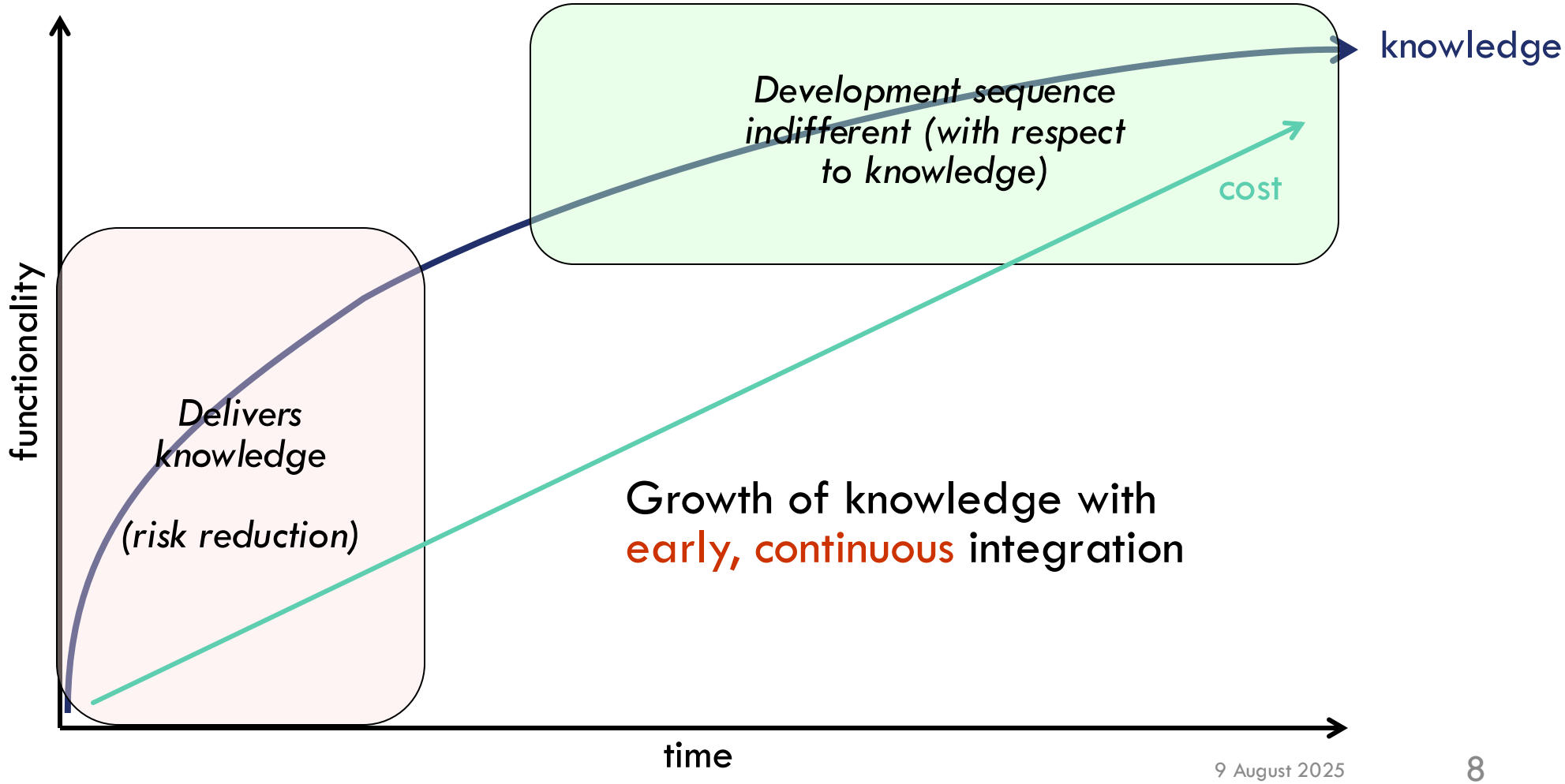
Growth of functionality and knowledge with big-bang integration



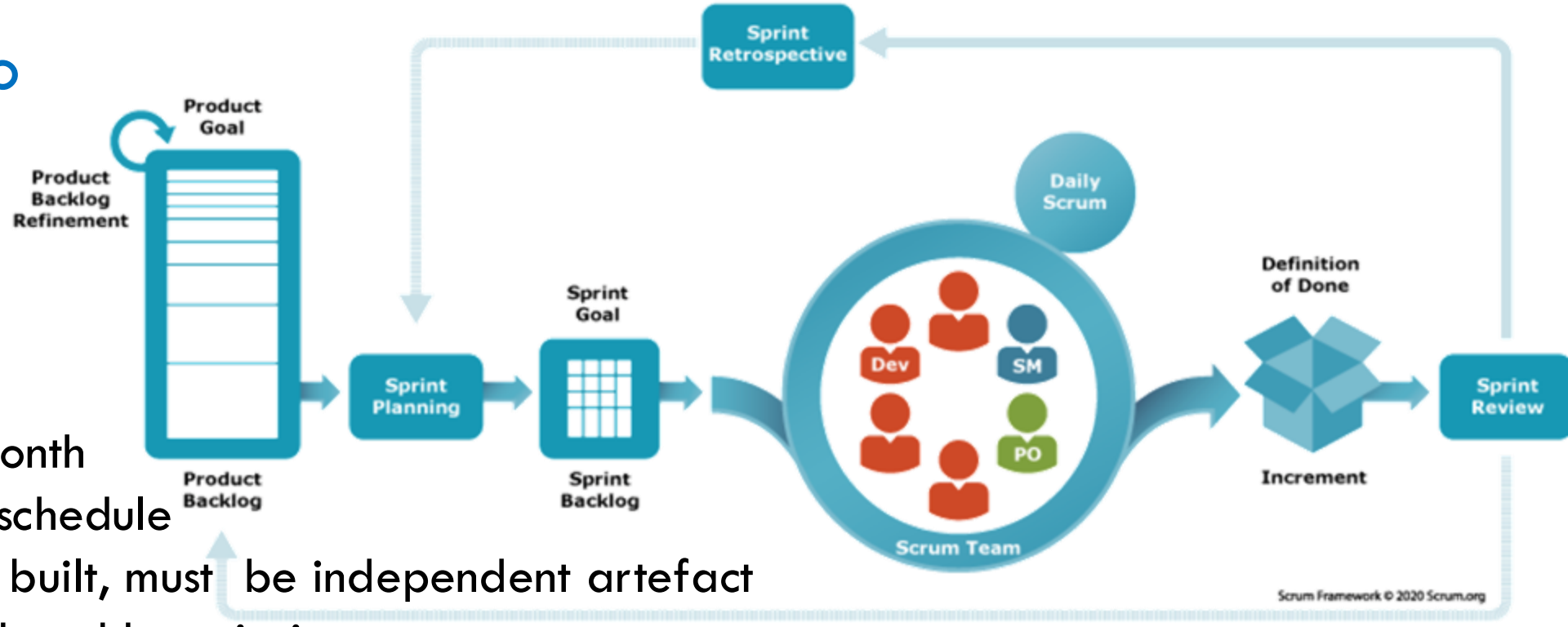
Scrum: Finished functionality

It pays to learn early in the project

Up front design only reduces risks if problem is well-understood



Scrum lingo



Goal: end objective

Sprint: fixed time ≤ 1 month

Planning: items to do + schedule

Product: Software to be built, must be independent artefact

Backlog: work items, ordered by priority

Team: ≤ 10 people, no sub teams, no hierarchy

Scrum master (SM): coach and mentor of the team, liaison to management

Daily scrum: 15 min meeting to discuss progress

Increment: outcome of sprint: concrete stepping stone toward the Product Goal

Definition of done: quality as agreed, work finished

Undone work: not finished, quality not OK

Sprint review: Inspect outcome + future adaptations

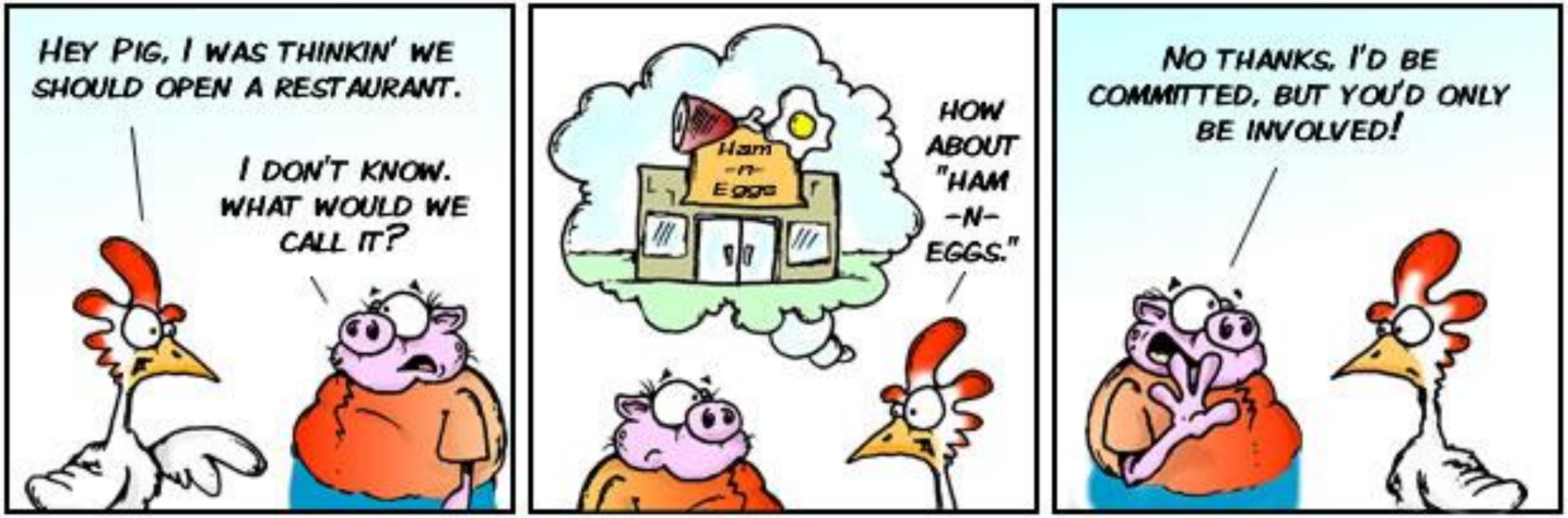
Sprint introspective: how to improve quality & effectiveness

Scrum who?

- Team of 10 people or less *(fewer is better)*
  - Need more people? => split product, make multiple Scrum teams

1 Scrum master (SM)	1 Product owner (PO)	*n-2* Developers
liaison Team ↔ org	can be shared by multiple teams	must have skills needed
guards quality	develop project goal	quality work: "definition of done"
removes obstacles to progress	plan project backlog + priority	sprint + daily planning
chairs scrum events	communicate with customers	training by Pair programming
not a leader, but a servant	provides funding	

Lingo: Pigs and chickens



- Scrum pigs: Team members (SM, PO, developers)
  - fully committed
- Scrum chickens: customers, users, vendors, management
  - involved: needed, but indirectly
- Scrum is about empowering the pigs and getting the chickens out of the kitchen

Scrum artifacts?

Artifacts represent either work or value

Each artifact contains a commitment to ensure it provides information that enhances transparency and focus against which progress can be measured:

	Product backlog	Sprint backlog	Increment
Commitment	Product goal	Sprint goal	Definition of done
What is it?	emergent, ordered list of needs to improve product	Set of product backlog items to work on	Concrete stepping stone towards product goal

	Product goal	Sprint goal	Definition of done
What is it?	Future state of product for team to plan sprints	Single objective for sprint	Definition of done
Characteristics	Clear and concise Transparent in backlog	Flexible in how achieved Makes ≥ 1 increments If impossible: cancel sprint	Formal definition of quality all devs commit to !quality => !done

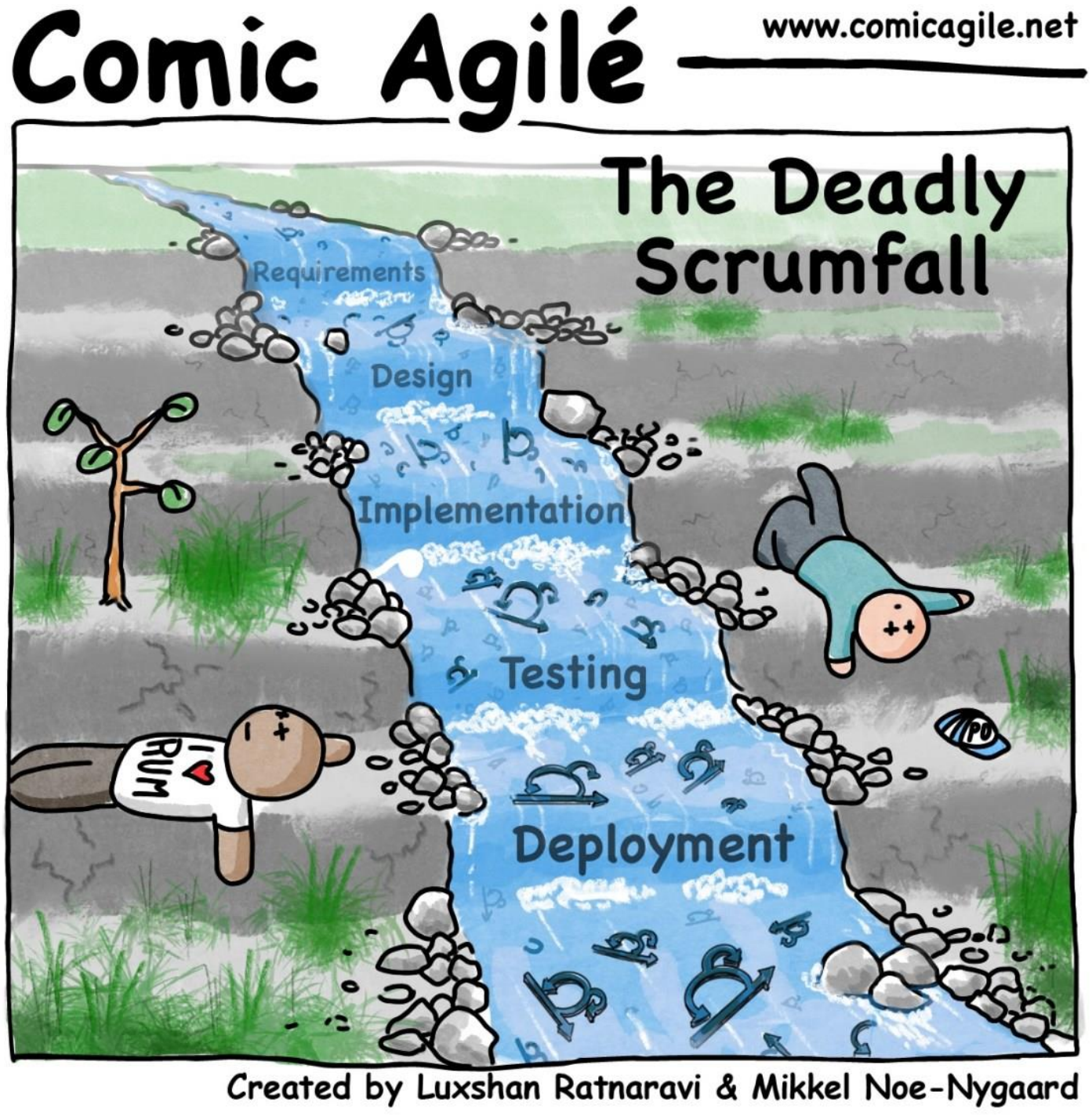
Scrum Events

- Events only happen inside a Sprint
- Event: formal opportunity to inspect and adapt Scrum artifacts
- All events at regular time and place
- Ideally no meetings outside events

Event type	Duration	Subject	Output	Participants
Sprint	≤ 1 month (2 weeks)		Increments	Mostly developers
Sprint planning	≤ 8 hours if 1 month	Product Backlog, Product Goal, Definition of Done	Sprint Backlog, Sprint Goal	Scrum Team
Sprint review	≤ 4 hours if 1 month	Increment, Sprint, Product Backlog, Progress toward Product Goal	Product Backlog	Scrum Team, Stakeholders
Sprint introspective	≤ 3 hours if 1 month	Sprint, Definition of Done	Actionable improvements, Definition of Done	Scrum Team
Daily scrum	15 mins daily	Progress toward Sprint Goal	Progress toward Sprint Goal	Developers

Scrum Team

- The team
  - looks at the requirements
  - considers the available technology
  - evaluates its own skills and capabilities
  - determines how to build the functionality, modifying its approach daily
  - figures out what needs to be done and selects the best way to do it.
- The heart of Scrum = *the iterations + creative process*



CSC3003S ASD09 Scrum

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Scrum Rules

- ScrumMaster = rule-master
- Rules = everyone knows how to play
  - if not enforced: people waste time figuring out what to do
  - If disputed: time is lost while everyone waits for resolution
- Rules work in thousands of successful projects
- To change rules, use Sprint retrospective meeting:
  - Changes should originate from the Team, not management
  - Changes only if the ScrumMaster is convinced
    everyone involved understands how Scrum works.
    - No rule changes until the ScrumMaster
      determines this stage has been reached

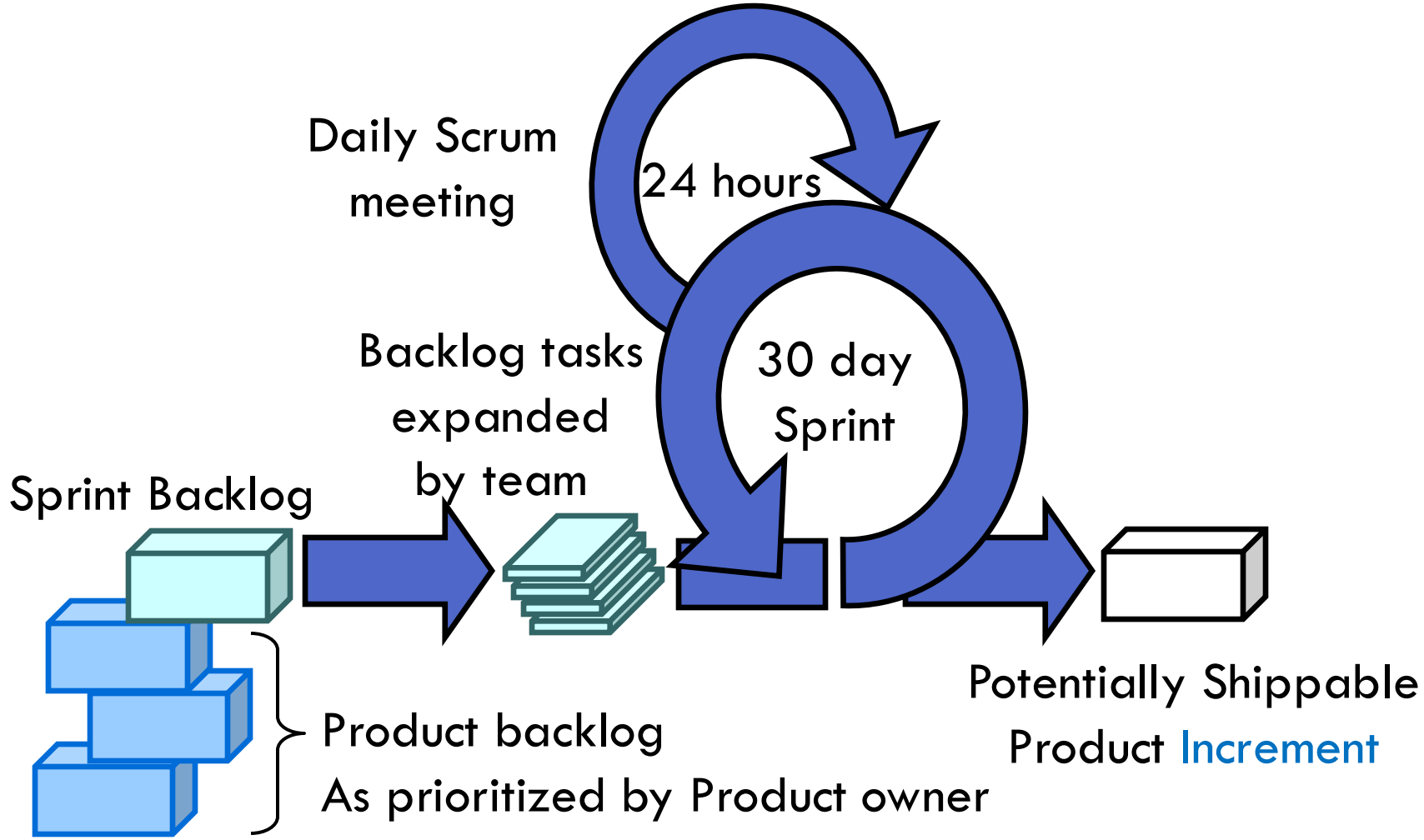
Scrum Flow

- Any project starts with a vision
  - In XP terms the “System Metaphor”
  - Leads to the logical architecture
- The Product Owner produces the Product Backlog
  - List of functional and non-functional requirements when turned into functionality deliver the vision.
  - Prioritized with the items most likely to generate value as top priority and divided into proposed releases.

Scrum Flow — Sprints

- Work is done in Sprints.
  - iteration of ≤ 20 consecutive workdays (month).
  - initiated with a Sprint planning meeting
    - the Product Owner and Team collaborate about what will be done for the next Sprint.
      - Select from the highest priority Product Backlog
      - Product Owner tells the Team what is desired,
      - Team tells the Product Owner how much it believes it can turn into functionality over the next Sprint.
    - ≤ 8 hours
      - Prevents hand-wringing about what is possible.
  - Goal is to get to work, not to think about working

Sprints


Daily Scrum
meeting
24 hours
Backlog tasks
expanded
by team
30 day
Sprint
Sprint Backlog
Potentially Shippable
Product Increment
Product backlog
As prioritized by Product owner



CSC3003S ASD09 Scrum

A. Intro to Scrum

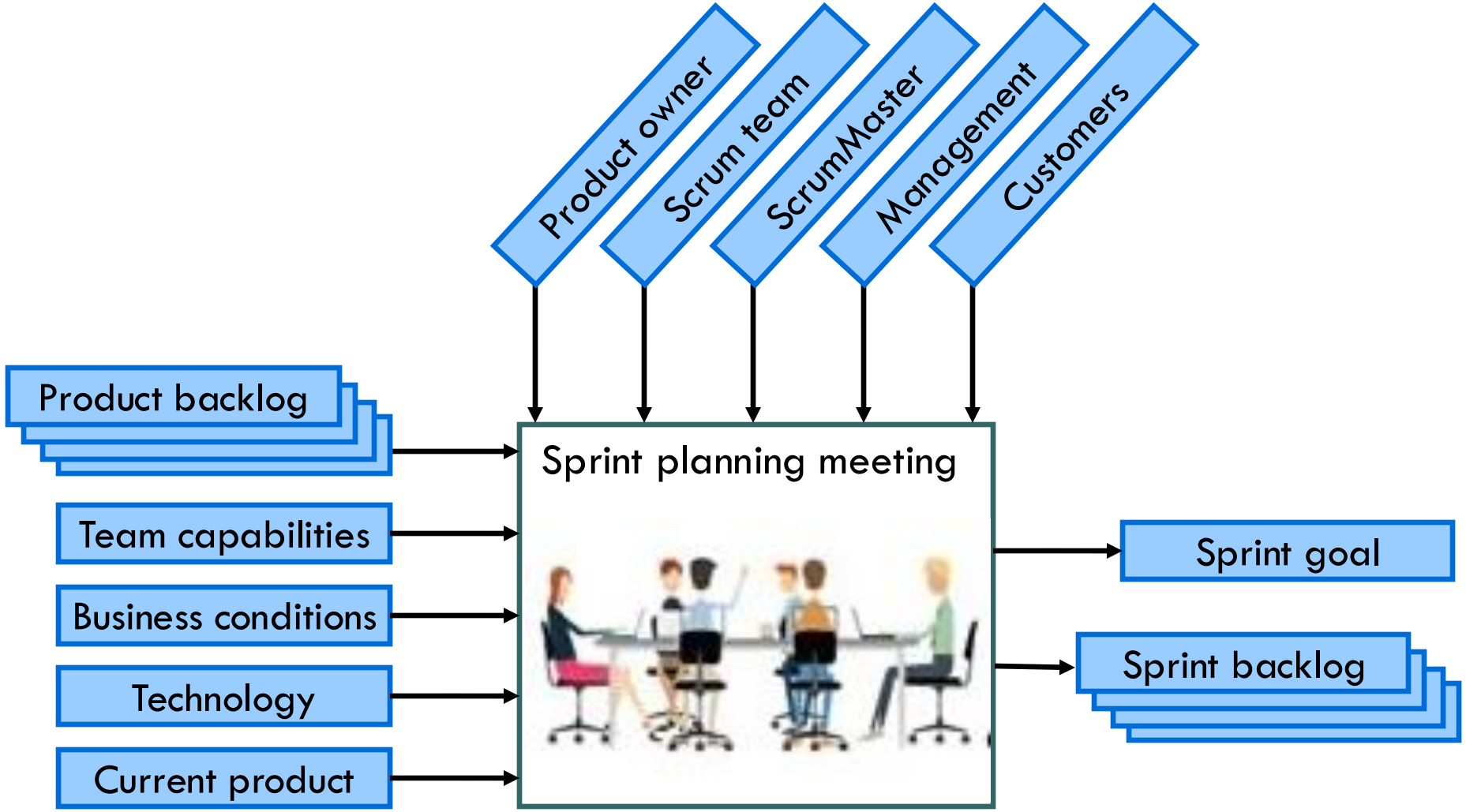
B. Scrum flow

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Sprint Planning Meeting ≤ 8 hours



2 segments ≤ 4 hours each:

4 hours: select Product backlog;

4 hours: prepare Sprint backlog.

Sprint planning meeting: 1st segment:

- Select Product Backlog items that it believes it can turn into an increment
- Team will demonstrate this increment to Product Owner and stakeholders at the Sprint review meeting after Sprint.
- Team makes suggestions, Product Backlog selection done by *Product Owner*.
- Team decides how many Product Backlogs to include in Sprint

Sprint planning meeting: 2nd segment:

- Team manages its own work
- Chosen tasks are placed in a Sprint Backlog
- Product Owner is available to answer questions from the Team
- Team works out how to turn the selected Product Backlog into increment.
- Others only observe or answer questions
- Output of the 2nd segment is the Sprint Backlog
  - tasks, time estimates, and assignments.
- Task list might not be complete
  - it must be complete enough to have commitment of all Team members
  - carries them through the first part of the Sprint,
  - while the Team devises more tasks in the Sprint Backlog.

Daily Scrum meeting

- Time-boxed to 15 minutes
  - No matter how many team members.
- Held in the same place at the same time every workday.
  - best is first thing in the day, so members think of what they did the day before and what they plan to do today.
- All team members must attend.
  - If a Team member can't attend, must attend by remotely or have another Team member report why
- Starts on time
  - ScrumMaster starts the meeting at the appointed time, regardless of who is present.
  - Any members who are late pay fine to the ScrumMaster immediately.

Daily Scrum Meeting

- ScrumMaster asks everyone 3 questions
  1. What did you do yesterday?
  2. What will you do today
  3. What stands in the way of getting your work done
- One answerer talk, other people listen
  - Short answers: $\frac{15\ mins}{3\ Qs * ppl\ present} \approx$ 30 seconds per answer
  - no digressions, discussions etc.
- Purpose
  - **Synchronize** work of all Team members daily
  - **Schedule** meetings needed to help progress
- Chickens 🐓 can attend, but cannot communicate, nonverbal or otherwise
  - chickens cannot initiate follow-up meetings with Pigs

Sprint: Time-Boxed

- Sprint is time-boxed ≤ 30 days (more typically 2 weeks)
  - time needed for a Team to build something of significant interest to the Product Owner and stakeholders and bring it to a state where it is potentially shippable.
  - maximum time to do work without artefacts and documentation to support thought processes.
  - maximum time that most stakeholders will wait
    - without losing interest in the Team's progress
    - without losing their belief that the Team is doing something meaningful.

Sprint

- During the Sprint
  - The Team *can* seek outside advice, help, information, and support.
  - *No unsolicited advice* to the Team.
  - The Team is self-managing.
- The Team has committed to the Product Backlog .
  - The Product Backlog is **frozen** until the end of the Sprint.

Changing Sprint Backlog

- If Team cannot complete Sprint Backlog
  - can consult with the Product Owner on which items to remove.
    - If too many items removed,
      ScrumMaster can terminate the Sprint

- If Team wants to do more Backlogs
  - Team asks Product Owner which
    Product Backlog items to add.

Non-Viable Sprint

- If the Sprint proves not to be viable, ScrumMaster may terminate the Sprint and initiate a new Sprint planning meeting.
  - The ScrumMaster can make this change if requested by the Team or Product Owner.

- Sprint can prove to be not viable
  - if the technology proves unworkable,
  - if the business conditions change so Sprint is not of value to the business,
  - if the Team is interfered with during the Sprint by anyone outside the Team.

Team Responsibilities During a Sprint

- The Team members have two administrative tasks during the Sprint:
  - *attend* the Daily Scrum meeting,
  - keep the **Sprint Backlog up-to-date** and available in a public folder on a public server, visible to all.
    - New tasks must be added to the Sprint Backlog as they are conceived, and the running, day-to-day estimated hours remaining for each task must be kept up-to-date.

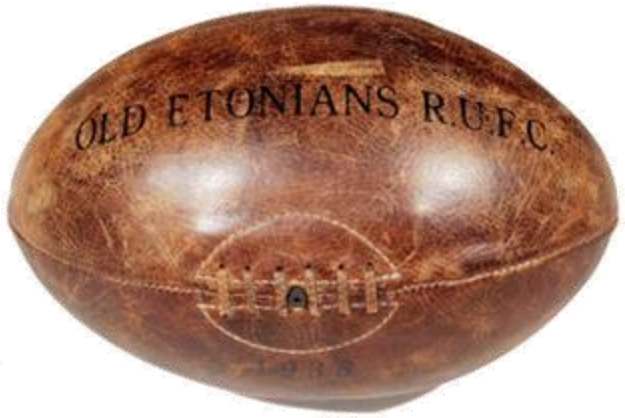
Sprint Review Meeting ≤ 4 hours

- The Team should spend ≤ 1 hour preparing
- Team presents increment(s) to Product Owner and Stakeholders
- "Definition of Done" means complete & can be shipped or implemented.
  - If “done” has another meaning
    make sure that the Product Owner and stakeholders understand it.

- Anything not 'done' cannot be presented
- Subjects:
  - Sprint goal, Sprint Backlog, Product Backlog completed
  - what went well and what not
  - answer stakeholder questions
  - stakeholder poll to get their impressions, needs, and priority of needs
  - product owner adds and reorders product backlogs

Sprint Retrospective Meeting ≤ 3 hours

- Attended only by Team, ScrumMaster, and Product Owner (optional)
- All Team members briefly answer 2 questions:
  - What went well during the last Sprint?
  - What could be improved in the next Sprint?
- ScrumMaster summaries the answers
- Then Team prioritizes the order of issues and talks about potential improvements.
- The ScrumMaster facilitates Team's search for improvement of Scrum process
- Actionable items can be added as high-priority non-functional Product Backlog items.



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Scrum's Artefacts

- Scrum has artefacts
  - Product Backlog
  - Sprint Backlog
  - Burndown Charts
- Can be managed using an Excel spreadsheet or cloud-based tools
  - Consider support for distributed teams

Product Backlog

- Prioritised list of first cut refinements.
- Product Owner is free to
  - adjust the order in which Product Backlog items are developed
  - add new items
- This brings in the concept of agility and change

Backlog item	Estimate (person-weeks)
Break props into separate rigid bodies	**1**
Arrow projectiles stick to environment	**0.5**
Ragdolls can lose limbs	**2**
NPCs avoid dynamite thrown at them	**4**
Improve exception handling	**0.5**

Sprint Backlog

- List of the things that will be “done” during the Sprint
- Each item has an estimate of how long it should take to complete, usually measured in hours.
- During the Sprint’s 30 day period, the Project Team must update the Sprint Backlog
- Keeping the Sprint Backlog updated is key:
  - not only does it allow us to work out how fast a team can work (their velocity), it is an early warning indicator.

Sprint Backlog Update

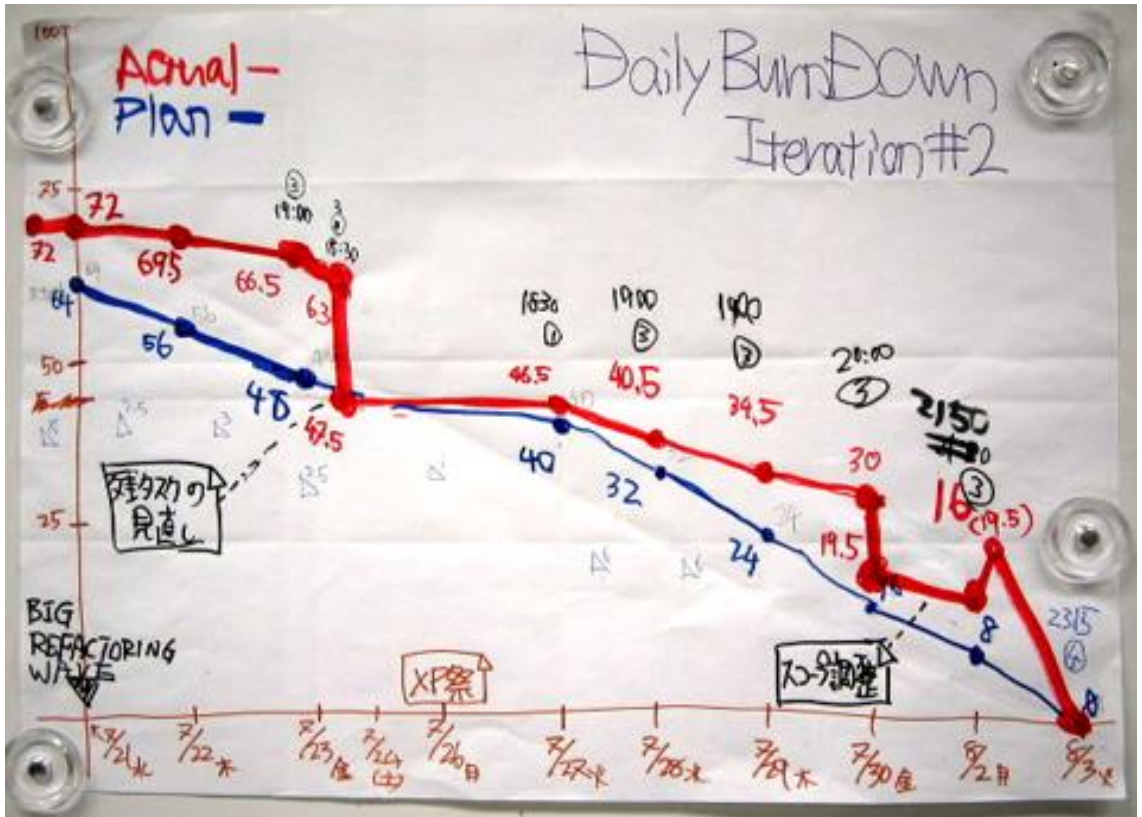
Sprint 1														
9/8/2006				Sprint Day	1	2	3	4	5			6	7	8	9
					Mo	Tu	Wec	Thur	Fri	Sat	Sun	Mon	Tue	Wed	Thur
				Hours	152	144	136	124	116			112	106	96	84
	19 days work in this sprint			Remaining											
ID	Backlog Item	Description	Owner	Estimate											
1	Minor	Remove bad formatting in login screen	GJ	8	8	4	2	0							
2	Minor	Correct legacy code in reorts	KJ	8	8	4	0								
3	Minor	Modify user file handling	CM	20	20	16	8	4	2			0			
4	Minor	Add network code for error cases	CM	40	40	36	32	30	26			24	20	20	18
5	Minor	Add company logos to all screens	DB	20	18	16	12	12	12			12	12	12	12
6	Major	Add algorithm to remove jitter	JG	56	56	56	56	56	56			56	56	56	56
7	Major	Change Design for workflow	DC	72	64	60	60	60	56			56	50	42	42
8	Super	Supervision/Guidance	CAM	32	32	28	28	24	24			22	20	20	16

Burndown Charts

- These are used to identify the rate that work is being completed.
- *Graphical representation of the sprint backlog*
- It illustrates the work done.
- Demonstrate a steady drive to zero hours remaining:
  - it represents a pace of work that should be sustainable.
  - however, some work takes longer than others,
  - some are even shorter,
  - may not be a perfect straight line.

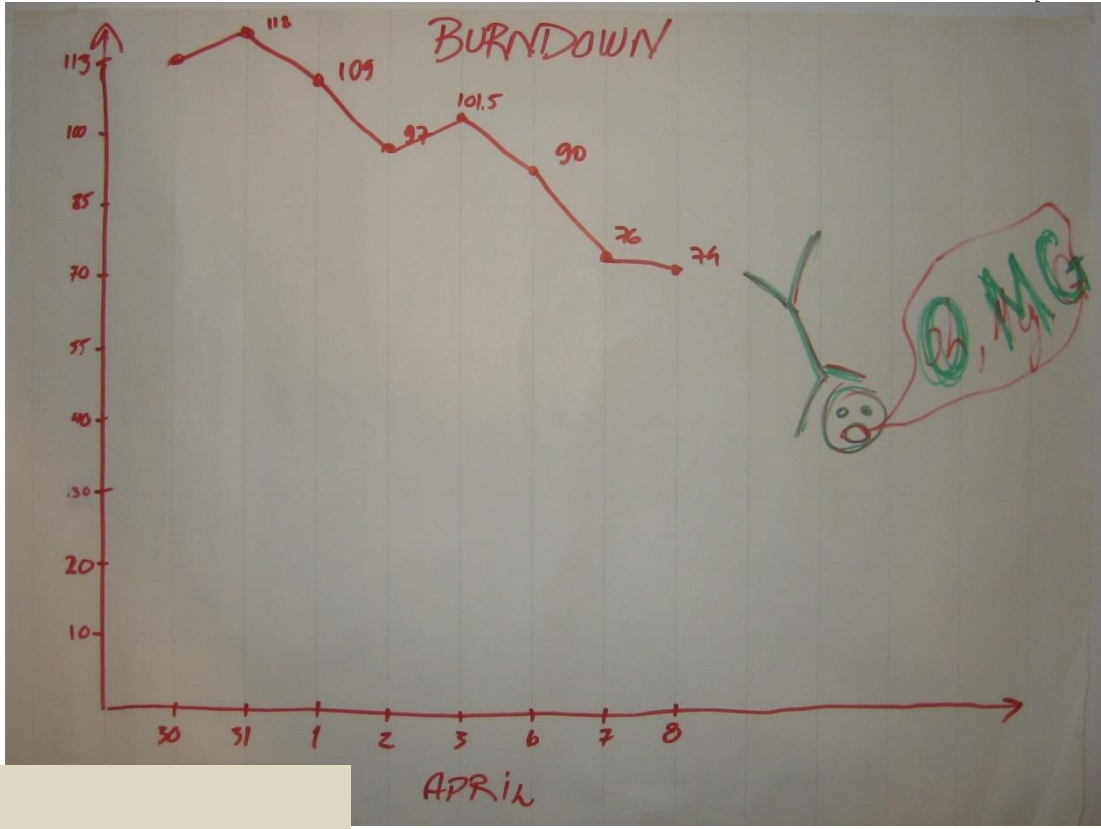
Burndown Chart

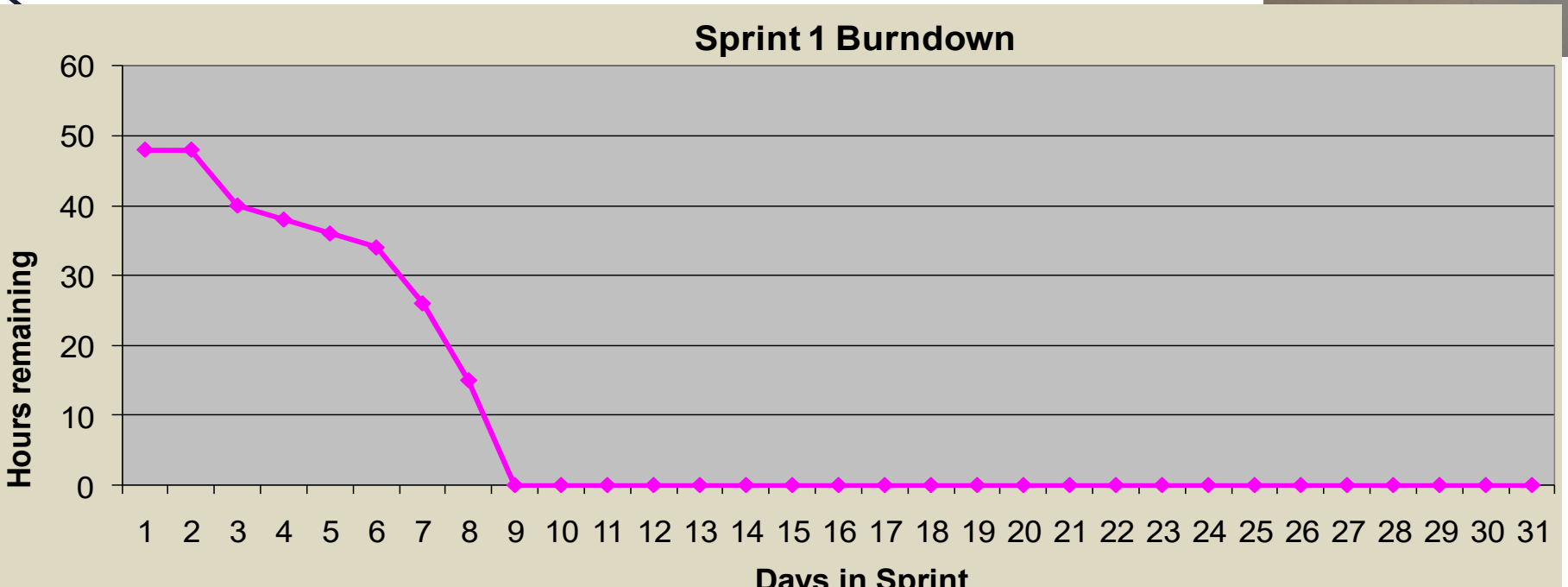
- Initially in a sprint there is no work done so the chart is as follows
- It is easy to see whether the work is being done in time.
- The following indicates it is not being done fast enough



Burndown Chart – example

- Too slow, due to:
  - inexperienced developers
  - hard roadblocks
  - distractions
  - bad planning





Or to fast



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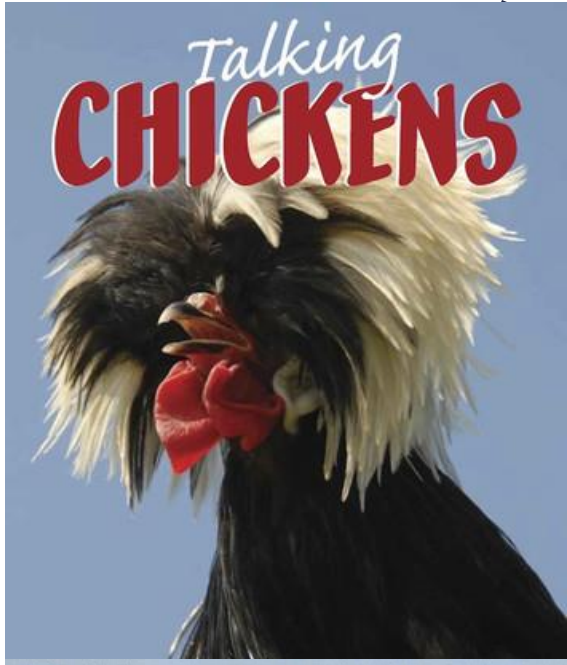
E. Scrum Problems

What can go wrong: *Loss of Rhythm*

- Sprints are not always the same length.
  - On a well-executed Scrum project the team establishes a natural rhythm.
  - Each sprint must be the same length.
  - Different length Scrums break rhythm
    - Sprints feel like arbitrary units of time selected more by outside forces not designed to improve productivity of the team.
    - Makes it hard for teams to plan

What can go wrong: Talking Chickens

- Chickens at daily Scrum ask questions or make observations.
  - Only pigs are allowed to speak
  - Talking chickens: slippery slope.
    If a chicken is allowed useful comment,
    how do we later prevent non useful talk?
  - Have to draw the line, best to be strict



What can go wrong: Missing pigs

- Not all pigs attend the daily Scrum meeting.
  - Some companies have flexible hours
  - Flex time, and the night work habit of developers,
    makes it common to have some members of a team late.



What can go wrong: *Missing Pigs II*

- it can be difficult for a team meet for a daily Scrum every day.
  - a daily meeting, at the same time in the same place each day help a project establish and maintain its rhythm.
  - If too many pigs miss daily Scrums too often then perhaps the meeting is too long or deviating from the 3 standard questions.
- When run well, daily Scrum ≤ 15 minutes and useful for every pig.
  - There can be specific deadlines that requires people to skip a daily Scrum
    - only in emergencies
  - when too many pigs miss meetings is there a problem

What can go wrong: *Specialized Job Roles*

- A project team has specialized job roles such as Architect, Designer, DBA, or Tester.
  - Scrum teams need to have a "*we're all in this together attitude*"
    - This can be undermined if a team has specialized job descriptions or roles.
  - It is simplistic to think that everyone can be a DBA and everyone can write server-side J2EE or .Net code.
- A successful Scrum team need not be all generalists.
  - Each specialist must accept responsibility for the system as a whole.

Wrong: *the Daily Scrum is for the ScrumMaster*

- The Daily Scrum is a status update from team to ScrumMaster.
  - ScrumMaster must be a servant
  - ScrumMaster takes notes about who committed to what work and why some other task wasn't completed.
  - Feels like status meetings
- Two main purposes of the daily Scrum
  - to provide a coordination to all on the project.
    - Everyone knows where everyone else is.
  - Each team member makes commitments in front of his peers.
  - If the commitment is not fulfilled, ScrumMaster's does not rebuke the team member.
    - They should feel bad enough.

Wrong: *ScrumMaster assigns work*

- Work is assigned by the ScrumMaster

  not signed up for by developers.

  - Self-organization is one of the underlying principles of Scrum.
  - If ScrumMaster assigns work, it undermines the responsibility of developers; they self-organize to achieve the goal.
  - Even occasional assignments by ScrumMaster are harmful
  - Teams must feel completely in control of their own work.